



Determining the refractive index of a plastic crescent using Snell's Law

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We used Snell's Law and the relationship between incident angles and refracted angles to determine the index of refraction of a crescent shaped piece of plastic. We did this using a laser and a rotation plate allowing us to measure the angles. We determined that the angle of refraction for the piece of plastic was 1.77 ± 0.02 , which falls into the acceptable range for indices.

I. INTRODUCTION

Exploring the world of optics, the study of light and how it interacts with other objects, through hands-on experimentation is a captivating way to understand some of the intricacies of the world. One aspect of optics that can be studied is refraction. Refraction is the bending of light as it crosses a boundary [1]. The way refraction can be measure is through the index of refraction, n . It is a dimensionless value that indicates the ability light has to bend in a medium.

It may not sound like the refraction has any impact on day to day life, but that couldn't be further from the truth. Think about how many people wear prescription glasses. In the United States over 60% of adults wear prescription glasses or contacts [2]. Otherwise consider a phone. The design of the camera has to account for the refraction of the lens. Understanding the refraction of a camera lens plays a large role in capturing clear and sharp images while taking a picture.

This paper looks at the experiment we performed on measuring the index of refraction of a crescent-moon shaped piece of plastic. We will do this by using a diode laser that shines light into our plastic crescent moon. From there we vary the angle of incidence that the light hits the plastic and take measurements of the refraction angle. We will be able to use our data graph the relationship between the two and determine the index of refraction.

II. FINDINGS

We start with the physics and mathematics behind refraction. We use Snell's law which can be seen below,

$$n_1 \sin \theta_1 = n_2 \sin \theta_2, \quad (1)$$

where n_1 is the index of refraction of the first medium, θ_1 is the incident angle of light, n_2 is the index of refraction of the second medium, and θ_2 is the angle of refraction. For the sake of our experiment we will use the refraction index of air for n_1 which is approximately 1.0. Therefore

we can ignore the value when calculating the refraction index of the plastic leaving us with,

$$\sin \theta_1 = n_2 \sin \theta_2. \quad (2)$$

Our experiment consisted of a diode laser than we aimed directly at our crescent-shaped plastic. The plastic was situated on a rotating disk that acted as a protractor, so we could measure the incident and refracted angles. We ran trials from normal to the plastic up to around 45 degrees for the incident angle. We were then able to plot the relationship. This plot can be seen below.

We can see that the relationship is linear even though the data tends away from linear near high angles on incident. We see that due to higher uncertainties in our measurements as well as the being relatively close to the critical angle. The critical angle is where light is no longer refracted and all of the light is reflected. We were able to fit a line to our data to determine the refraction index of our plastic crescent. The equation for line of best fit

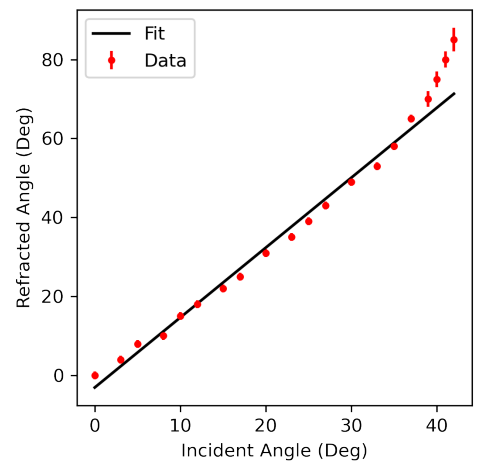


FIG. 1. The graph shows the relationship between the incident angle and the refracted angle.

was determined to be,

$$\theta_2 = 1.77(\theta_1) - 3.09, \quad (3)$$

where θ_2 is the refracted angle measured in degrees and θ_1 is the incident angle. We can use this line to determine our refraction index very simply. Our refraction index for the plastic crescent is the slope of our line. That means our refracted index is 1.77 ± 0.02 . The calculation for our uncertainty value can be found in Appendix A. We compared the index of refraction we found, 1.77, to an experiment rather previously by one of our colleagues team. The value they measured for the refracted index was 1.7. This allowed us to sanity check our work and determine that we were in the right ballpark with our measurements.

III. CONCLUSION

We were able to determine the value of the refractive index of a crescent-shaped piece of plastic in this lab by finding the relationship between the angle of incidence and the angle of refraction. A relationship which can be described by Snell's Law. We found through our collection and analysis that the index of refraction is 1.77 ± 0.02 . Our process could be improved through a more accurate way to measure the incident and refracted angle; especially near the critical angle near an incident angle of 45 degrees. A possibility to extend this experiment could lie in the efficiency the refraction index could be measured. This could be done by taking a collection of random mediums and running the experiment over and matching the measured refractive index to a list of given refractive indexes to determine what the medium is made out of.

Appendix A: Uncertainty Analysis

We estimated the uncertainties for the measurements of the angles to be one degree for all incident angles and one for all angles of refraction of less than 70 degrees. We had to increase the uncertainty to two and three degrees at and above 70 degrees because the angle was harder to get an exact measure of.

To calculate our uncertainty in the index of refraction we did a least-square analysis using Python. The block of code we used can be seen below.

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```
def least_squares(x,y,dy):
    w = 1/(dy**2)
    W = sum(1/(dy**2))

    mux = np.average(x,weights=w)
    varx = np.average((x-mux)**2,weights=w)
    muy = np.average(y,weights=w)

    # Least-squares form of m,b
    m = np.average((x-mux)*(y-muy),weights=w)/varx
    b = muy - m*mux

    # Least squares uncertainty
    db = sqrt((1+(mux**2/varx))/W)
    dm = 1/sqrt(varx*W)
    return m,b,dm,db
```

FIG. 2. This is the code from python we used to calculate our uncertainty in refraction index.

[1] S. Beeson, J. W. Mayer, S. Beeson, and J. W. Mayer, Patterns of Light: Chasing the Spectrum from Aristotle

to LEDs , 33 (2008).

[2] W. P. Team, How many people wear glasses? (2023).